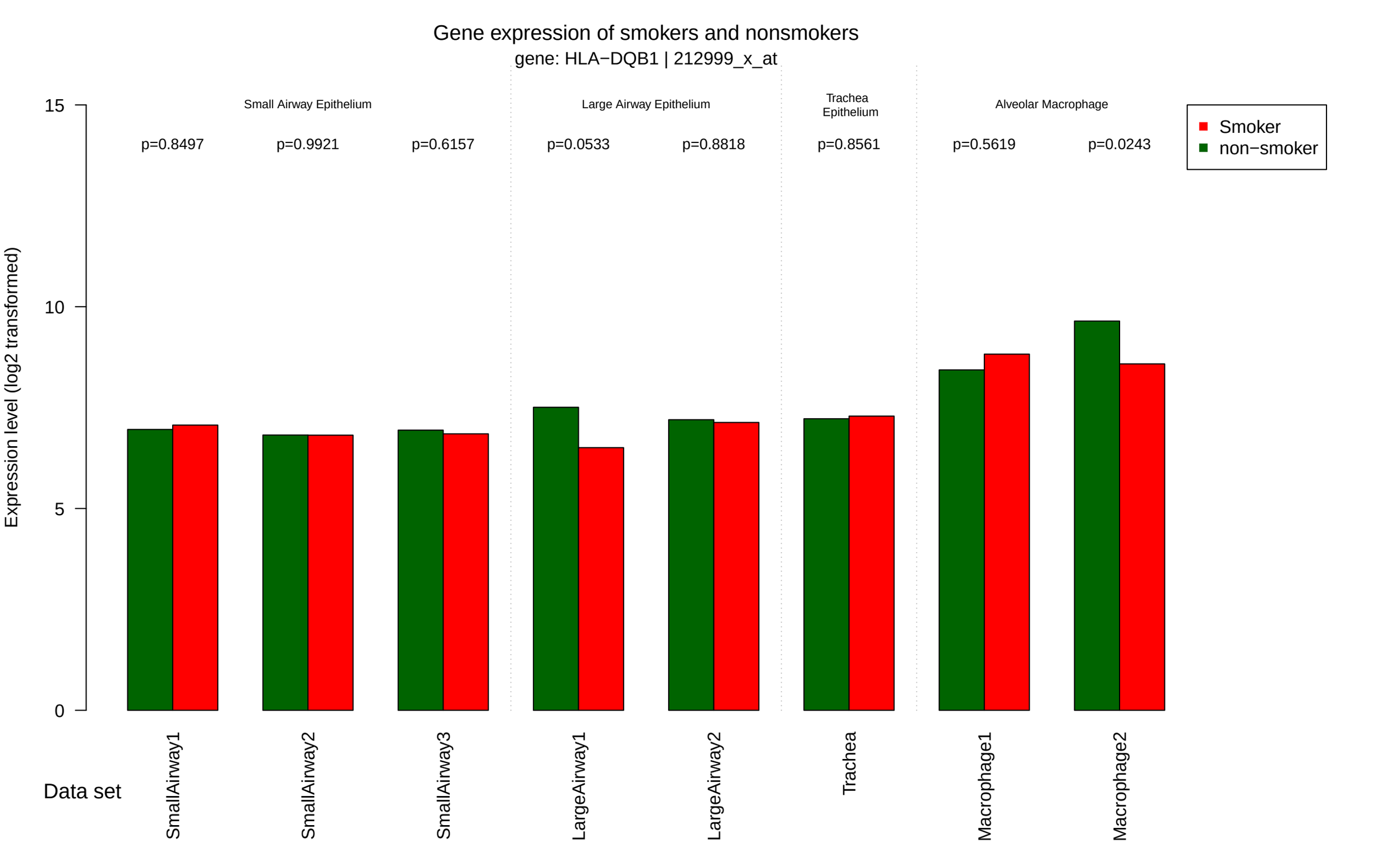
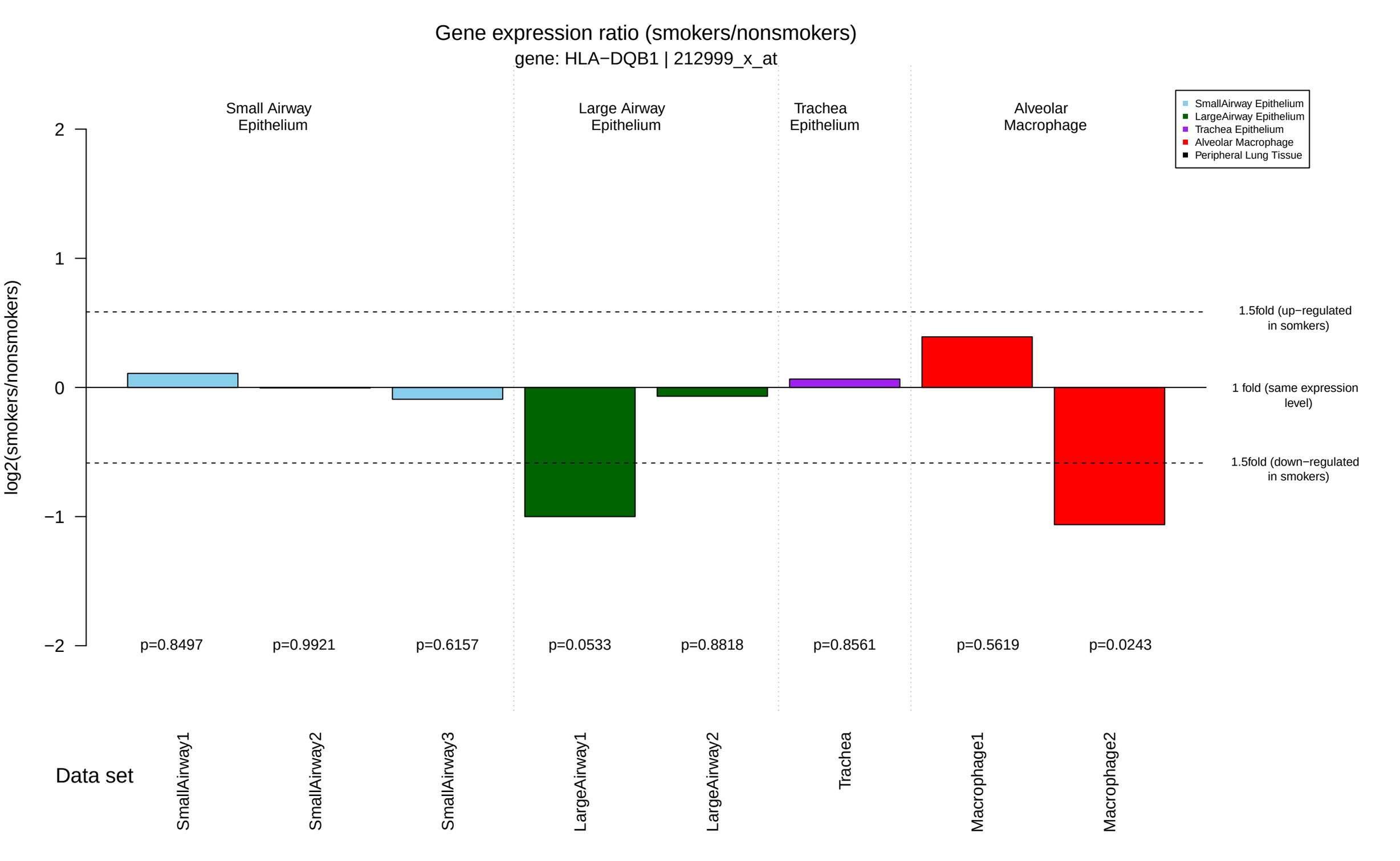
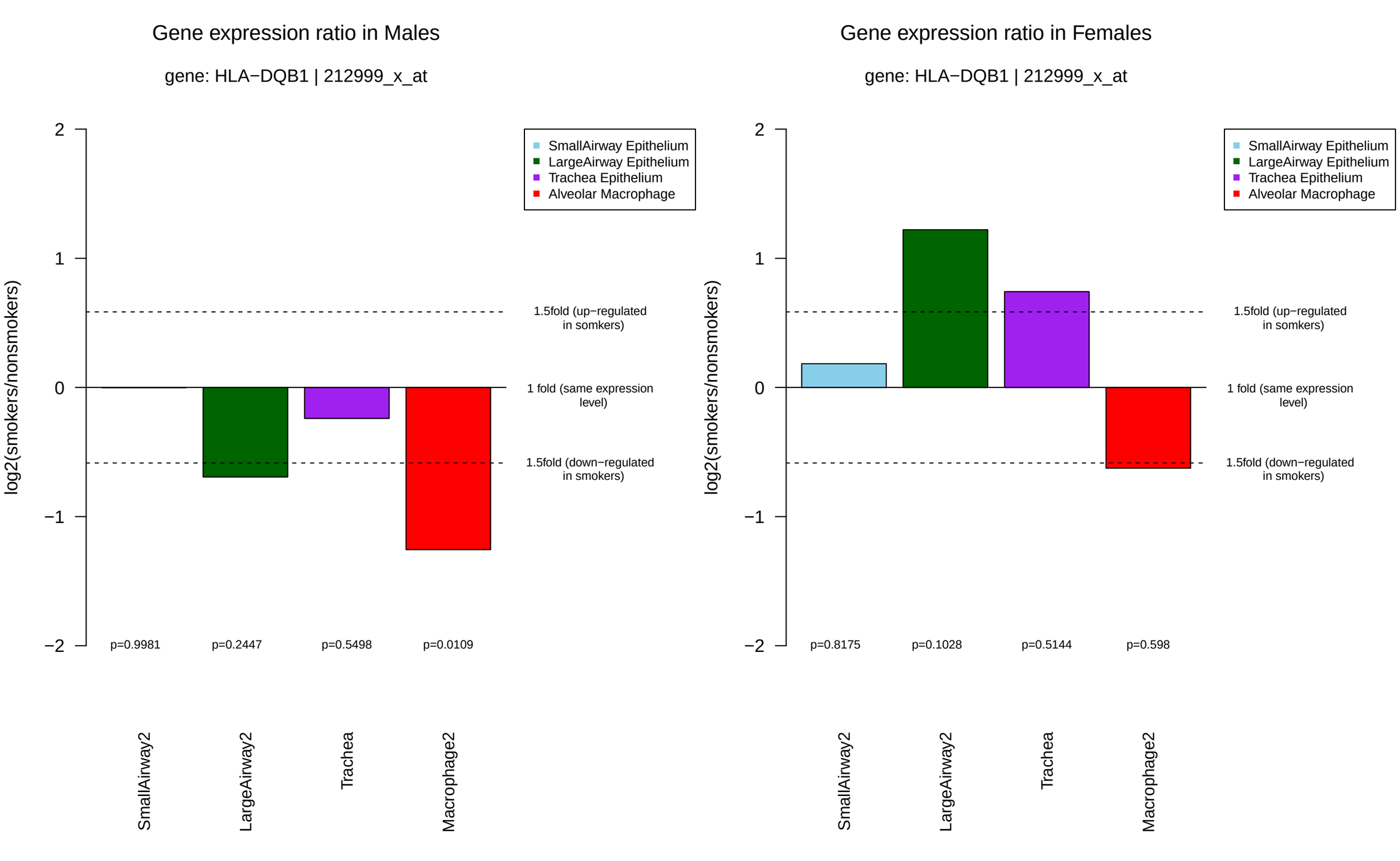
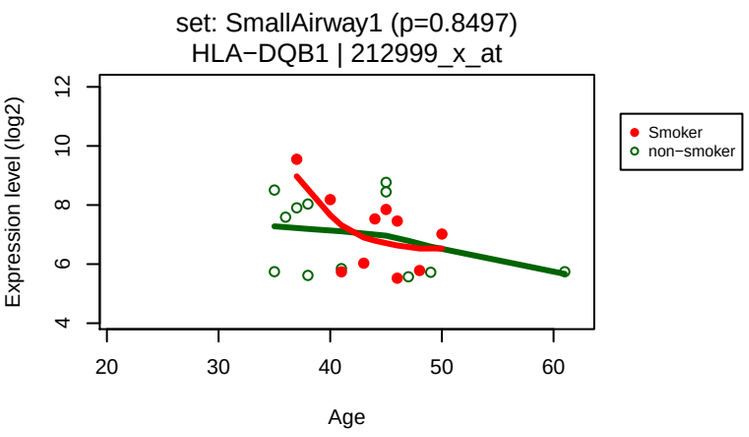
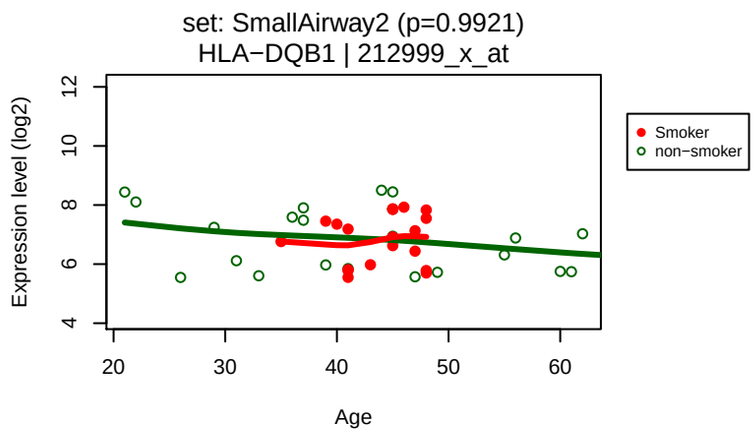
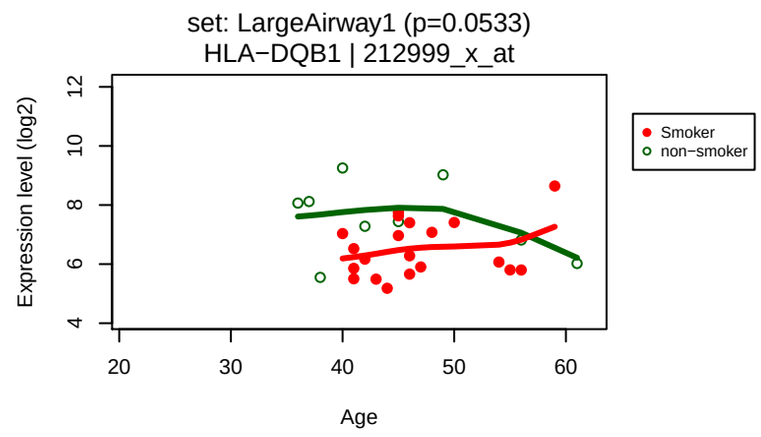

Gene expression of smokers and nonsmokers
gene: HLA−DQB1 | 212999_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8497
p=0.9921
p=0.6157
p=0.0533
p=0.8818
p=0.8561
p=0.5619
p=0.0243
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

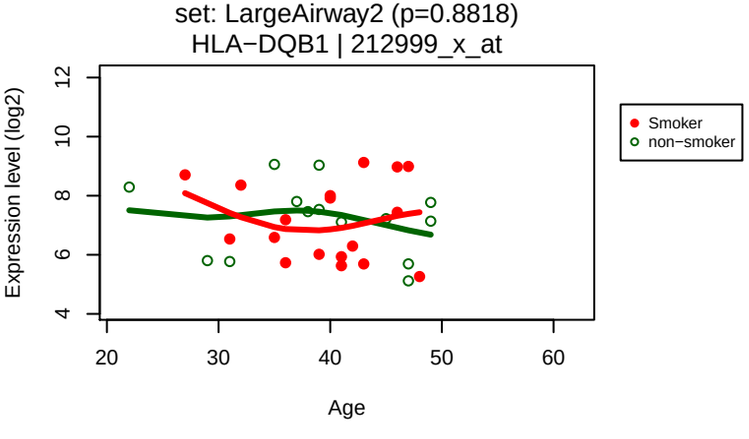

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQB1 | 212999_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8497
p=0.9921
p=0.6157
p=0.0533
p=0.8818
p=0.8561
p=0.5619
p=0.0243
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

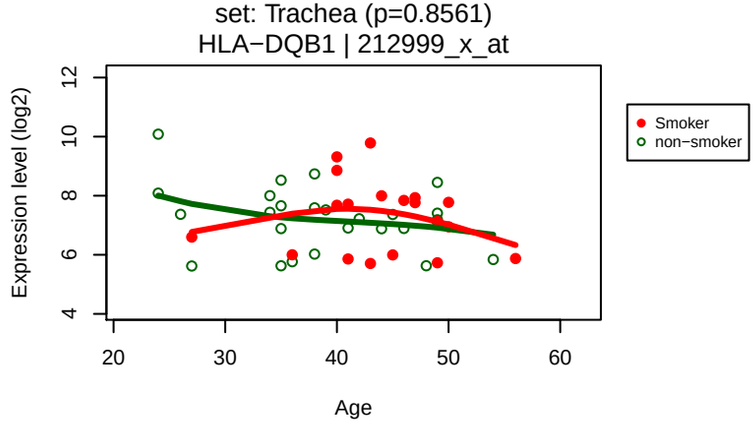

Gene expression ratio in Males
gene: HLA−DQB1 | 212999_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9981
p=0.2447
p=0.5498
p=0.0109
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQB1 | 212999_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8175
p=0.1028
p=0.5144
p=0.598
SmallAirway2
LargeAirway2
Trachea
Macrophage2

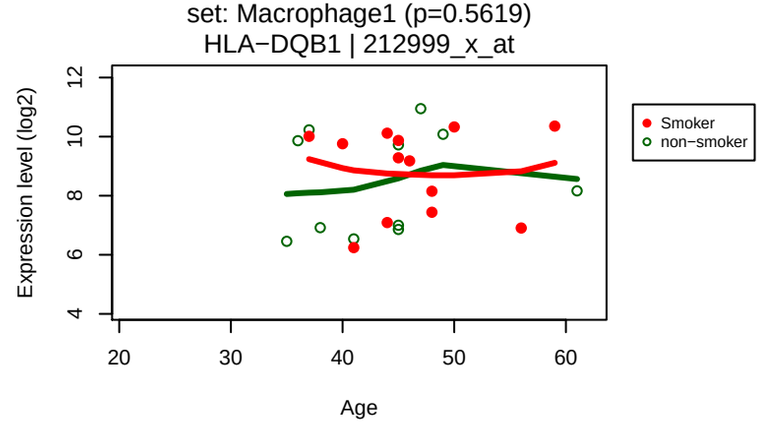

set: SmallAirway1 (p=0.8497)
HLA−DQB1 | 212999_x_at
Expression level (log2)
Age
Smoker
non−smoker

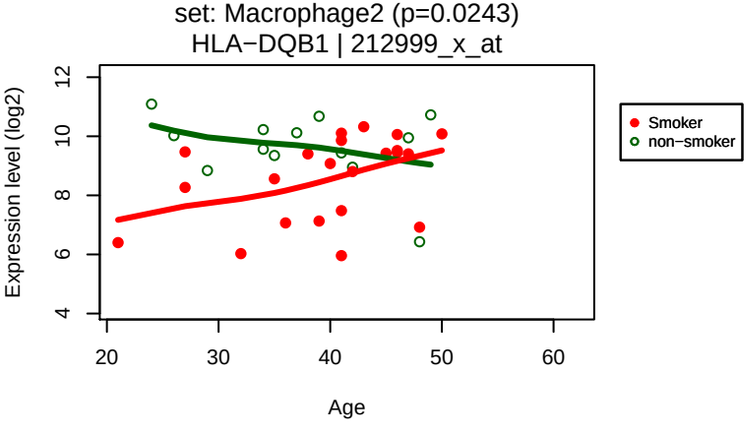

set: SmallAirway2 (p=0.9921)
HLA−DQB1 | 212999_x_at
Expression level (log2)
Age
Smoker
non−smoker

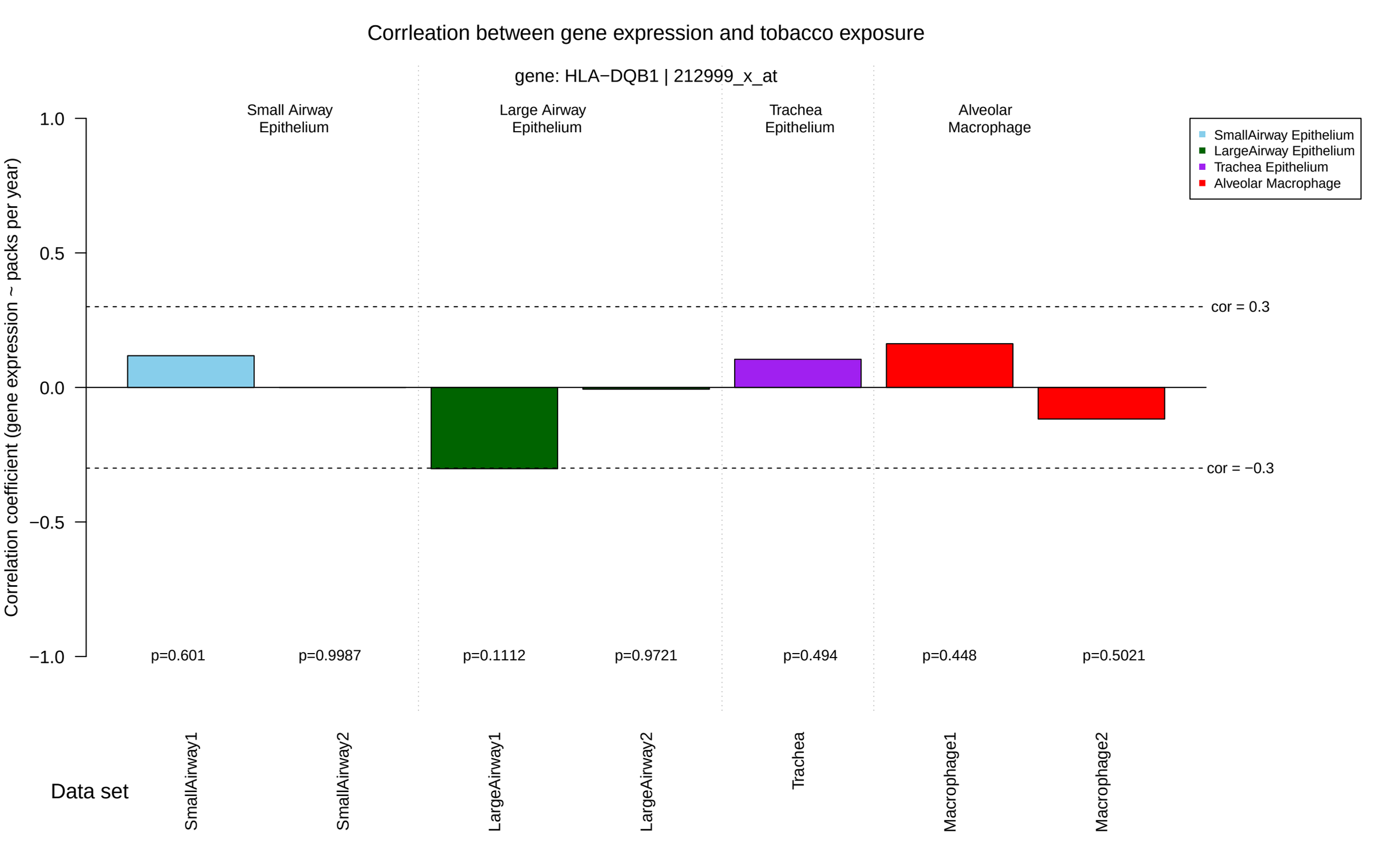

set: LargeAirway1 (p=0.0533)
HLA−DQB1 | 212999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8818)
HLA−DQB1 | 212999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8561)
HLA−DQB1 | 212999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5619)
HLA−DQB1 | 212999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0243)
HLA−DQB1 | 212999_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQB1 | 212999_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.601
p=0.9987
p=0.1112
p=0.9721
p=0.494
p=0.448
p=0.5021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2